

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: linearized DNA used for stable transfections.
Species Used	Mouse, SP-2, myeloma [Sp2/0-Ag14].		

Before the Pulse

Cell Growth Medium	DMEM, 10% Fetal Calf Serum (FCS) (GIBCO/BRL, Sigma)	Growth Phase at Harvest	50 to 70% confluency
		Pre-pulse Incubation	4° C, 10 min.

Wash Solution Wash two times in electroporation buffer

The Pulse **Instruments Used** Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature		
Electroporation Medium*	Phosphate Buffered Saline	Cuvette Gap	0.4 cm
Cell Density	5 x 10 (5) cells/pulse, stable transfection	Voltage	0.180 kV
Volume of Cells	0.5 ml	Field Strength	0.45 kV/cm
DNA Concentration	10 µg / pulse		
DNA Resuspension Buffer	Not given; pulse volume: 0.8 ml	Capacitor	960 µF
Volume of DNA	Not given; pulse volume: 0.8 ml	Resistor	(Pulse Controller) Ω none
After the Pulse		Time Constant	24 msec
Outgrowth Medium	DMEM, 10% Fetal Calf Serum (FCS)		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Note: Stable transfections generally do not use carrier DNA. Also, the level of selective agent required to kill off non-transfected cells needs to be established before transfection. The level required should kill non-transfected cells in approximately 7 days.

Outgrowth Temperature	37 °C
Length of Incubation	48 to 72 hrs.
Selection Method or Assay Used	G418 (stable transfections)
Electroporation Efficiency	Not given
Per Cent Survival	about 50%

Name of Submitter Jackie Beall

Institution Address University of Adelaide
Department of Biochemistry
P.O. Box 498
Adelaide 5001
Australia

Survey Number
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